



December 2025

(For CSE)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from Unit – I & Unit – II each and **One** full question from Unit – III.

20 Marks

- 1-



10. In process synchronization, which problem involves readers and writers trying to access a shared database?
 A) Producer-consumer problem B) Dining philosophers problem
 C) Readers-writers problem D) Banker's algorithm
11. 8. What does deadlock avoidance involve?
 A) Detecting deadlocks after they occur B) Allocating resources dynamically based on current availability
 C) Swapping processes to avoid deadlocks D) Killing processes to avoid deadlocks
12. What is the main purpose of process synchronization?
 A) To prevent processes from entering the critical section B) To avoid deadlocks
 C) To ensure that multiple processes access shared resources correctly D) To enable process communication
13. What is a mutex lock?
 A) A software-based synchronization technique B) A hardware-based memory management technique
 C) A type of semaphore D) A scheduling algorithm
14. What is the key idea behind the Banker's algorithm for deadlock avoidance?
 A) Processes must wait for resources in a circular manner B) Resources are allocated dynamically to prevent deadlocks
 C) All resources must be pre-allocated to processes D) It ensures the system always remains in a safe state
15. In deadlock detection, what is the resource allocation graph used for?
 A) To track process execution times B) To determine if a deadlock has occurred
 C) To allocate memory dynamically D) To swap processes between memory and disk
16. What is the main purpose of the paging mechanism?
 A) To increase system speed B) To prevent deadlocks
 C) To efficiently manage memory allocation D) To synchronize process communication
17. Which of the following page replacement algorithms is considered to be the most optimal, but not typically implementable in practice?
 A) First-In, First-Out (FIFO) B) Least Recently Used (LRU)
 C) Optimal Page Replacement D) Clock Page Replacement
18. In the Least Recently Used (LRU) page replacement algorithm, which page is replaced?
 A) The page that has been in memory the longest. B) The page that was used most recently.
 C) The page that has been least frequently accessed. D) The page that has not been used for the longest time.
19. Which of the following file allocation methods involves storing all blocks of a file together on contiguous blocks of disk?
 A) Linked allocation B) Contiguous allocation
 C) Indexed allocation D) Hash-based allocation
20. Which disk scheduling algorithm selects the request that is closest to the current head position?
 A) First-Come, First-Served (FCFS) B) Shortest Seek Time First (SSTF)
 C) SCAN D) C-SCAN

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a) Define Process. Explain the different parts of a process with a neat diagram.		4	L*1	1	2

- b) Apply the Preemptive Priority Scheduling algorithm to determine the average waiting time and average turnaround time for the given set of processes, where each process is defined by its arrival time and burst time (in milliseconds), and a higher numerical value denotes a higher priority.

Process	Arrival Time	Burst Time	Priority
P1	0	4	1
P2	1	3	2
P3	2	1	3
P4	3	5	4
P5	4	2	4

6 L3 2 3

- c) Outline with a neat diagram the different services provided by an Operating System. From these, discuss any two user-specific services and any two system-specific services briefly.

6 L1 1 2

2. a) Outline the structure of a Process Control Block (PCB) with the help of a neat diagram. List its major components and briefly explain any two of them.
- b) Apply the First Come First Served (FCFS) scheduling algorithm to compute the average waiting time and average turnaround time for the given set of processes. The table below provides their arrival times and burst times (in milliseconds).

4 L1 1 2

Process	Arrival Time	Burst Time
P1	2	2
P2	5	6
P3	0	4
P4	0	7
P5	7	4

6 L3 2 3

- c) Explain Multilevel Queue Scheduling with the help of a suitable example and diagram.

6 L2 1 2

3. a) List the different Multithreaded Models in Operating Systems. Explain any one model in detail with the help of a neat diagram.
- b) Explain the different Scheduling Queues used in an Operating System. Illustrate your answer with a neat queuing diagram.
- c) Apply the Round Robin (RR) scheduling algorithm with a time quantum of 2 ms to calculate the average waiting time and average turnaround time for the given set of processes. The table below provides their arrival times and burst times (in milliseconds).

4 L2 2 2

6 L2 1 2

Process	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

6 L3 2 3

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | List and explain two types of semaphores based on their usage. | 4 | L2 | 3 | 2 |
| b) | Outline the basic method of paging in detail with the help of a neat diagram | 6 | L2 | 4 | 2 |
| c) | What is a deadlock? Explain the <i>four necessary</i> conditions that must be satisfied simultaneously for a deadlock to occur. | 6 | L2 | 3 | 2 |
| 5. a) | Calculate the number of bits in the logical address and physical address when the logical address space consists of 16 pages of 512 words each, mapped to a physical memory of 64 frames. | 4 | L3 | 4 | 2 |
| b) | Discuss Peterson's solution to the critical section problem for two processes along with the pseudocode. | 6 | L2 | 3 | 2 |
| c) | Outline the pseudocode of the Safety Algorithm and the Resource Request Algorithm used for deadlock avoidance. | 6 | L2 | 3 | 2 |
| 6. a) | Explain the working of Resource Allocation Graph (RAG) algorithm for deadlock avoidance. | 4 | L2 | 3 | 2 |
| b) | Explain paging mechanism with Translation Look aside Buffer (TLB) | 6 | L2 | 4 | 2 |
| c) | Discuss the solution to the Readers-Writers problem to achieve synchronization. | 6 | L2 | 3 | 2 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Discuss the steps involved in handling a page fault with a neat diagram. | 8 | L2 | 5 | 2 |
| b) | Apply the SCAN and C-SCAN disk scheduling algorithm for the following sequence to compute seek length and average seek length with an initial position of disk head at 76
17, 29, 53, 76, 124, 137, 201, 269 | 8 | L3 | 5 | 2 |
| 8. a) | Explain the concept of free space management in operating systems. List the different types of free space management techniques used and explain any two techniques in detail. | 8 | L2 | 5 | 2 |
| b) | Discuss the Optimal Page Replacement and Least Recently Used (LRU) Page Replacement techniques. Then, apply both algorithms to the following page reference string and calculate the total number of page faults in each case. Assume the system has three frames.
2,3,1,5,2,4,3,5,2 | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
